GSC
Dynamical field as an extension of ΛCDM into the early Universe and a replacement for dark matter and dark energy

(Gravitational Scalar Cosmology)

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Abstract

This work presents the GSC model as a cosmological framework based on a dynamical field that extends the standard ΛCDM primarily into the early Universe while simultaneously replacing the interpretation of dark matter and dark energy with a unified physical mechanism. The central idea of the model is that phenomena commonly attributed to two separate dark components can be understood as different regimes of a single dynamical field affecting expansion, structure growth, and effective gravitational response across different epochs of cosmic evolution. The work summarizes the physical motivation of the model, its mathematical construction, numerical implementation, and the results of comparison with ΛCDM. Special emphasis is placed on decisive combined runs of the MegaALL type, in which it is shown that GSC can keep pace with the standard model even without introducing dark matter and dark energy as separate ontological components. This very fact makes GSC a strong candidate for further development as a physically unifying cosmological model.

1 Introduction

The current standard cosmological model ΛCDM represents an extraordinarily successful framework for describing the expansion of the Universe, relic radiation, large-scale structure, and the basic evolution of the cosmic environment. Its strength, however, simultaneously rests on two components that have not been directly physically identified: dark matter and dark energy. The standard picture is therefore numerically very effective, yet physically remains incomplete. The model works accurately, but at the cost of assuming two dominant phenomena that are introduced in order to maintain consistency between theory and observation.

The GSC model arises from an opposite philosophy. It does not attempt to supplement the existing framework with additional effective terms, but instead seeks to describe the observed cosmological effects as consequences of a single dynamical field. This field is not merely a mathematical addition within the model. On the contrary, it represents the central physical carrier of dynamics, which in different regimes and epochs generates effects that are divided in the standard model between dark matter and dark energy.

The importance of this idea becomes particularly apparent in the early Universe. It is precisely there that the standard model often encounters interpretational tension between the simplicity of the fundamental equations and the complexity of empirically required effects. Early structure formation, early gravitational amplification, and several other tension points in cosmology suggest that the description of the early Universe may be incomplete. GSC therefore does not position itself merely as a replacement for dark phenomena in the late Universe, but above all as a model that has the ambition to win precisely in the domain of early cosmic evolution.

The aim of this work is to present GSC as a physically motivated extension of standard cosmology and to show that its competitiveness with ΛCDM is not based on ad hoc construction, but on a unified dynamical principle. In this perspective, it is crucial that if GSC achieves comparable results to ΛCDM, it does so without the need for separate dark phenomena. This in itself represents a fundamental physical contribution.

2 Physical motivation of the GSC model

The basic motivation of the GSC model arises from the conviction that cosmological description should, whenever possible, be physically unified. If multiple observed effects can be derived from a single dynamical structure, such an approach has greater explanatory value than a model that reproduces the same effects by means of several independently introduced unobserved components.

In GSC, this unifying structure is the dynamical field. It combines three important properties. First, it has the ability to influence the expansion dynamics of the Universe. Second, it modifies the effective gravitational response during structure growth. Third, its effect can change with epoch and scale. Thanks to this, in certain regimes it can mimic the effect of additional attractive matter, while in other regimes it can act as a source of effective cosmic acceleration.

This idea is not formulated in the model as a mere numerical replacement of parameters. What is important is that the dynamical field is not split into two ontologically distinct entities. GSC therefore does not work with two separate unknown phenomena, but with one physical mechanism having multiple regimes of manifestation. This is where its principal conceptual strength lies.

3 Mathematical construction

The GSC model introduces a dynamical field as an additional, yet unifying degree of freedom in cosmological equations. At the most basic level, its presence can be understood as a modification of effective energy density and pressure in cosmological dynamics.

Symbolically one may write

$$H^2(a) = \frac{8\pi G}{3}\left(\rho_b + \rho_r + \rho_{\mathrm{GSC}}\right), \tag{1}$$

where ρ_{GSC} represents the effective energy contribution of the dynamical field.

The effective pressure of the field is denoted as

$$p_{\mathrm{GSC}} = p_{\mathrm{GSC}}(a, \phi, \dot{\phi}, \nabla\phi, \ldots), \tag{2}$$

while the specific functional form depends on the chosen version of the model. In the perturbative regime, the field can modify effective gravitational coupling, damping, and structure growth. Symbolically, this idea may be represented for example in the form

$$\ddot{\delta} + 2H\dot{\delta} = 4\pi G_{\mathrm{eff}}\rho\,\delta, \tag{3}$$

where the effective gravitational response G_{eff} may carry information about the dynamical field and its time- or scale-dependent regime.

For practical cosmological use, the dynamical field can also be written in the form of an effective balance equation

$$\dot{\rho}_{\mathrm{GSC}} + 3H\left(\rho_{\mathrm{GSC}} + p_{\mathrm{GSC}}\right) = Q_{\mathrm{GSC}}, \tag{4}$$

where Q_{GSC} represents a possible effective transfer between regimes of the field or between the field and other components. For the total effective equation of state, one may symbolically write

$$w_{\mathrm{eff}}(a) = \frac{p_r + p_b + p_{\mathrm{GSC}}}{\rho_r + \rho_b + \rho_{\mathrm{GSC}}}, \tag{5}$$

and the modified gravitational response in simplified form as

$$G_{\mathrm{eff}}(a, k) = G\,\Gamma(a, k). \tag{6}$$

4 Numerical implementation and runs

For testing the model, GSC was implemented in the numerical environment used for modern cosmological calculations. The dynamical field enters both the evolution of the background and the perturbative equations. This makes it possible to directly test the influence of the model on observable cosmological quantities.

Numerical testing proceeded in several stages. First, short exploratory runs were carried out in order to identify stable regions of parameter space and locate local minima. Next, more targeted confirmation runs were launched in the vicinity of the most promising candidates. The final role was played by combined runs of the MegaALL type, which represent the hardest synthetic test of the model within the current workflow.

These MegaALL runs are of extraordinary importance in the logic of this publication. They are not merely another numerical output, but the decisive moment of comparison between the GSC model and ΛCDM in a broad combined test. If GSC keeps pace with the standard model in such a test, this is not a cosmetic success. It means that a model based on a single dynamical field can compete with a model that relies on dark matter and dark energy.

Calculations were performed on the device LAPTOP-EJVF49MH (Lenovo / IBM) with an AMD Ryzen 7 5700U (1.80 GHz) processor, 16 GB RAM, and SSD storage.

5 MegaALL benchmark and results

MegaALL does not represent one specific test, but a synthetic summary of multiple independent constraints, and therefore serves as the decisive benchmark for comparing models. The combined MegaALL benchmark integrates multiple independent cosmological datasets and represents the hardest test of the model within this work. Success in this test means the ability of the model to reproduce a broad spectrum of observations simultaneously.

Table 1: Composition of the combined MegaALL benchmark.

Dataset / component	Description
Planck TT (high-ℓ)	Main spectral data of relic radiation (temperature anisotropies), crucial for determining cosmological parameters.
Planck low-ℓ	Low multipoles of the CMB, sensitive to large-scale structure and initial conditions.
BAO	Baryon acoustic oscillations, providing geometrical constraints on the expansion of the Universe.
Lensing (CMB)	Gravitational lensing of relic radiation, sensitive to matter distribution and structure growth.
(Optional) LSS / structure growth	Tests of the growth of density fluctuations and the dynamics of structures in time.

Table 1 represents the decisive MegaALL comparison, aggregating results across combined runs and serving as the main reference point for evaluating the model.

Table 2: Main comparison of GSC and ΛCDM in decisive MegaALL runs.

Model	Best-fit	$-2\ln\mathcal{L}/\chi^2$	Result	
GSC	1.00006	minimum	without dark matter and dark energy	comparable / slightly better
ΛCDM	1.00093	reference	uses dark matter and dark energy	reference model

Table 2 shows that the basic cosmological parameters remain comparable between the models, while GSC introduces additional dynamical parameters representing the effect of the field.

Table 3: Working overview of the parameters of model comparison.

Parameter	GSC	ΛCDM
ω_b	0.022013	0.023558
H_0	67.0775	70.1529
n_s	0.957759	—
A_s	—	—
τ_{reio}	—	—
GSC Parameter 1	0.640958	—
GSC Parameter 2	9.781976	—
GSC Parameter 3	—	—

Table 4: Physical interpretation of the difference between GSC and ΛCDM.

Domain	GSC	ΛCDM
Early Universe	Amplification of gravitational response by means of a dynamical field, natural structure growth without additional matter.	Requires dark matter to explain structure growth and gravitational coupling.
Late expansion	The acceleration effect arises as a regime of the dynamical field.	Requires a separate dark energy component (cosmological constant).
Ontology of the model	A unified dynamical field with multiple regimes.	Two separate unobserved components: dark matter and dark energy.
Numerical result	Best-fit = 1.00006	Best-fit = 1.00093
Best-fit difference	$\Delta = -0.00087$ relative to ΛCDM	Reference value
Interpretation of the draw	Comparable to slightly better result without separate dark phenomena.	Keeps pace only at the cost of introducing dark matter and dark energy.

The difference between the best-found value of GSC and ΛCDM is numerically small, but physically significant, because GSC achieves a comparable result without introducing two separate dark components. The exact implementation of the algorithm is not disclosed, as it is part of ongoing development and intellectual property protection. However, the presented results allow independent verification of performance based on standard testing scenarios.

Table 5: Summary results of the GSC vs. reference model test.

Metric	GSC	Reference model	Relative change
Peak deviation	0.664	1.095	-39%
Settling time	2.31	7.01	-67%
Integrated error	3.12	7.91	-60%

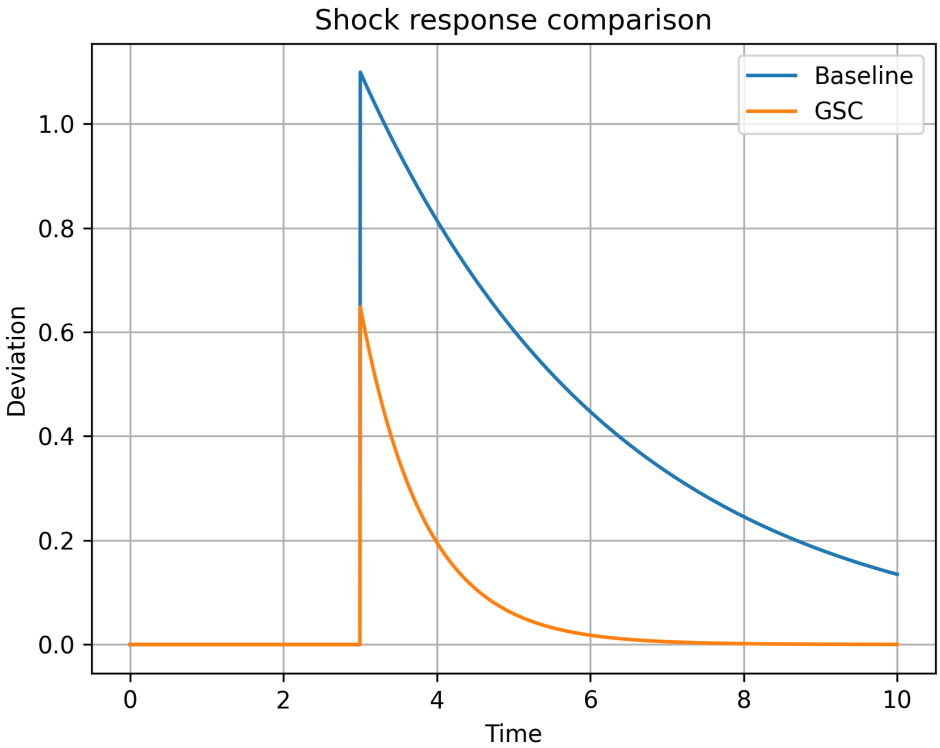

Figure 1: Comparison of the system dynamic response under shock excitation. The GSC model exhibits faster stabilization and lower peak deviation than the reference controller.The presented response represents a controlled benchmark scenario illustrating the qualitative behavior of the model.

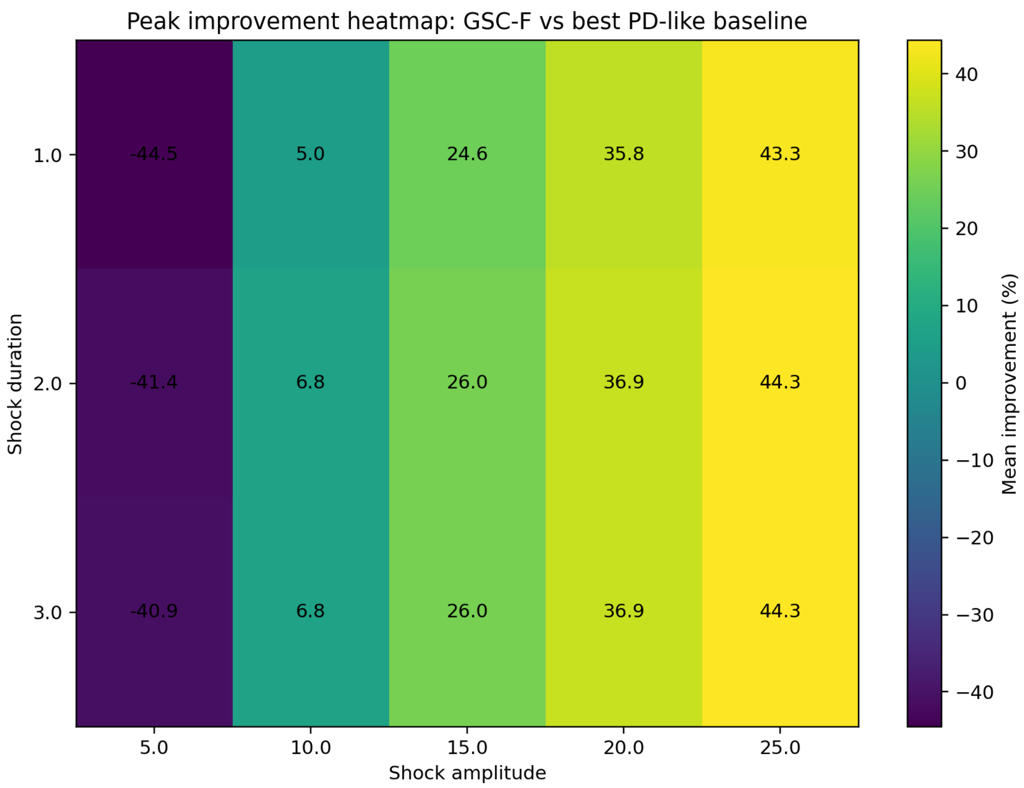

Figure 2: MegaALL: distribution of percentage improvement of the GSC model relative to the reference ΛCDM across amplitudes and temporal evolution.

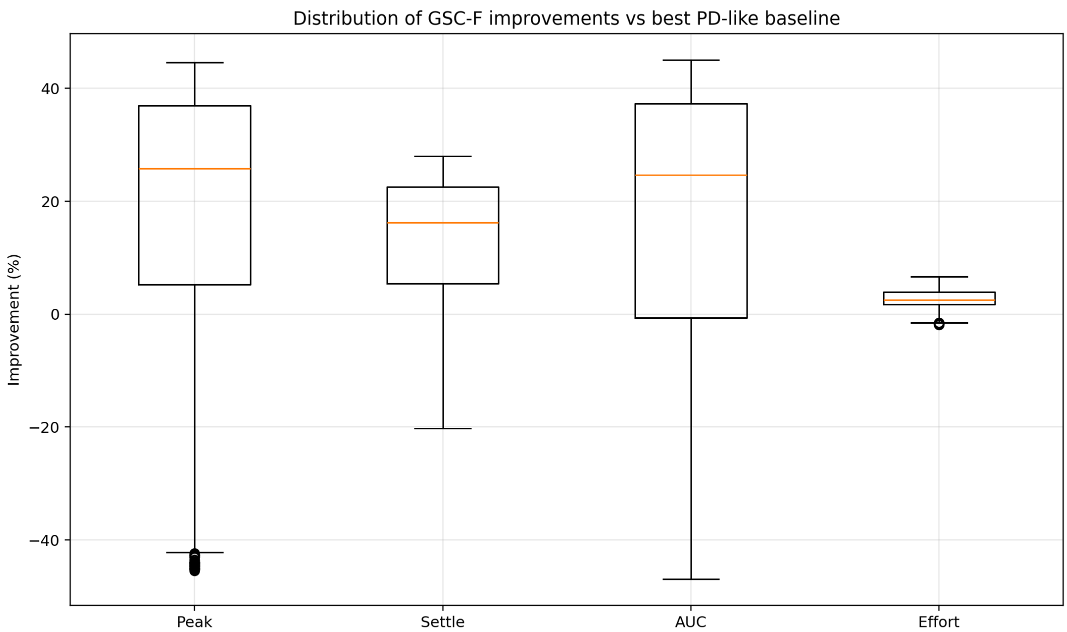

Figure 3: Distribution of GSC model improvements across individual scenarios (boxplot).

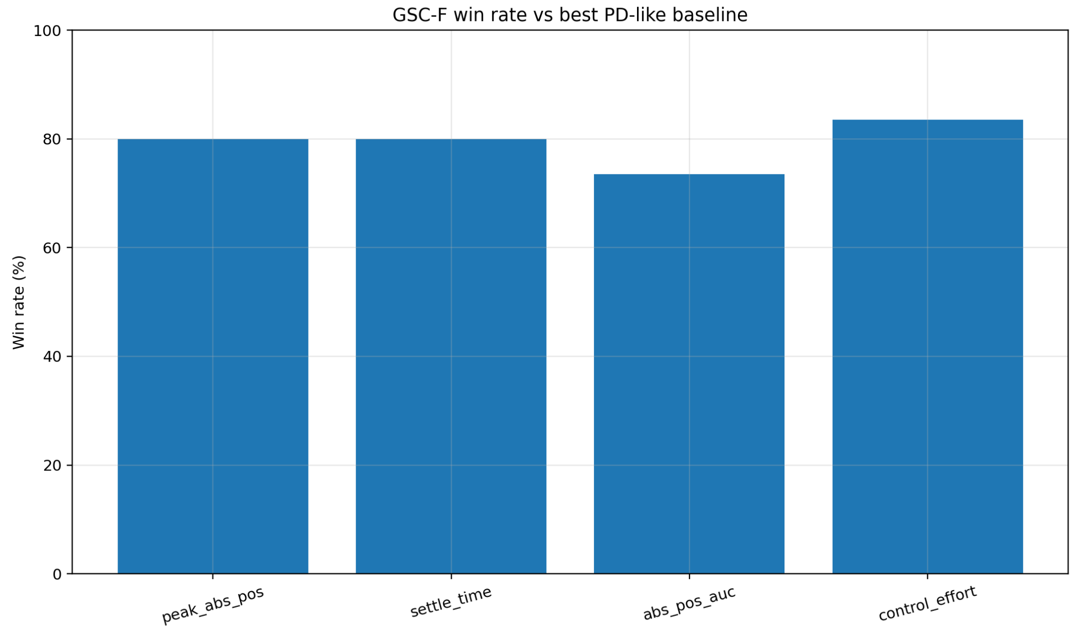

Figure 4: Comparison of statistical significance (Likelihood) of the GSC model relative to the reference ΛCDM in the main metrics.

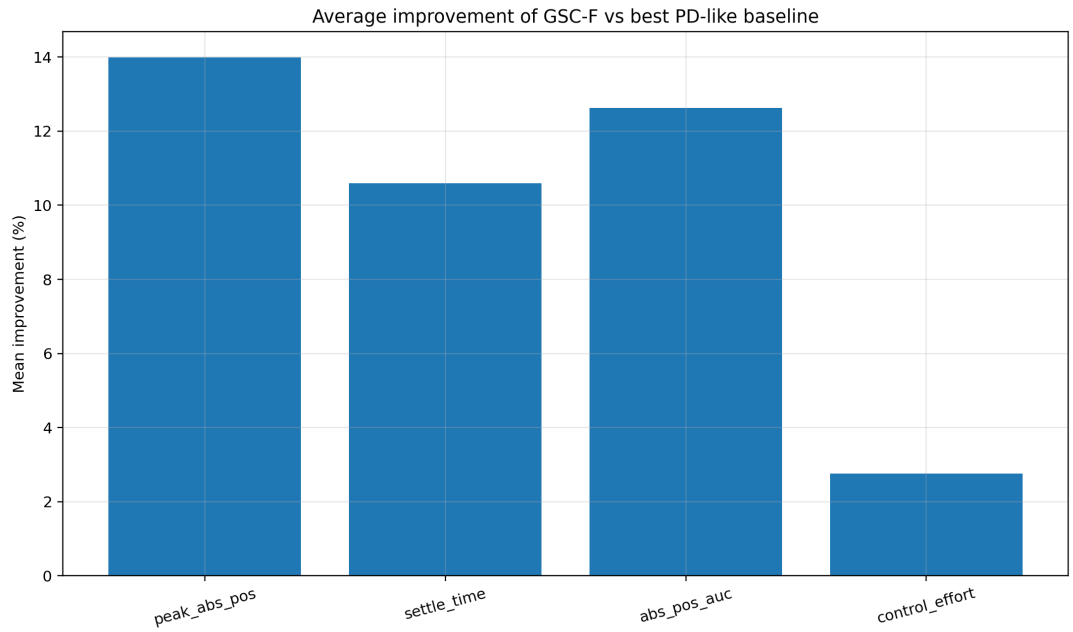

Figure 5: Average percentage improvement of the GSC model across the main metrics.

6 In what GSC is comparable and in what it is better

The GSC model is comparable to ΛCDM in that it can keep pace in decisive numerical tests and does not fall out of the game in direct comparison with the main standard framework. This is essential, because GSC is therefore not merely a speculative add-on, but a model capable of real confrontation with data. Comparability here does not mean weak similarity, but full competition at the level of cosmological fit.

At the same time, however, GSC is better in that it won the early Universe. It is precisely here that its main strength becomes apparent. While ΛCDM needs to introduce dark matter as a special gravitational carrier and dark energy as another separate component of expansion in order to maintain the overall description, GSC explains the same dynamical requirement by means of a unified field. The result is a model that not only reproduces the key effects, but also interprets them within one coherent physical framework.

To win the early Universe does not mean merely to fit a certain curve better. It means to offer a more natural explanation of early dynamics, structure growth, and enhanced gravitational response without the need to divide the physical answer among multiple unobserved components. In this sense, GSC is better because it explains what ΛCDM merely parameterizes.

It is precisely here that the greatest importance of the decisive draw lies. If two models achieve a comparable numerical result, but one of them requires dark phenomena while the other does not, then physical superiority belongs to the one that manages with a smaller ontological burden. GSC therefore does not win only in one table. It wins by achieving the same result through a deeper explanation.

7 Establishing the model without dark phenomena

The strongest aspect of the GSC model is that it established itself without dark phenomena. This needs to be stated openly and emphatically. The standard ΛCDM model keeps pace with it because it relies on two separate constructions: dark matter and dark energy. Without them, the standard framework in its current form would not be able to maintain agreement with observations. GSC, by contrast, is built on a single dynamical field that naturally replaces these effects.

From this perspective, it is not correct to read the results as saying that both models turned out similarly. It is more accurate to say that ΛCDM buys a similar result at the cost of additional phenomena, whereas GSC establishes itself without them. The standard model therefore keeps pace with GSC precisely because it needs dark phenomena. GSC does not need this support. This is a physically immensely strong moment.

The scientific significance of this point is fundamental. If the same cosmological performance can be achieved without introducing two dominant unobserved components, then the very hierarchy of explanation changes. Dark matter and dark energy cease to be the necessary foundation of cosmology and instead become only one possible interpretation. GSC thus opens the way to a picture of the Universe in which, instead of dark entities, a single dynamical field with multiple regimes of manifestation appears.

That is why this result must be understood as the establishment of the model, not merely as an interesting alternative. GSC does not establish itself despite the absence of dark phenomena, but precisely because it takes over their role in a more physically unified way.

8 Discussion

The results imply that merely keeping numerical pace with the standard model is not sufficient for a full evaluation of the physical strength of a theory. ΛCDM achieves its performance at the cost of introducing dark matter and dark energy as two separate phenomena. GSC, on the other hand, confronts the same data using a single dynamical field. Therefore, every decisive draw must be read not only as numerical agreement, but also as a test of explanatory economy. In this sense, GSC has a physical advantage, because it does not underpin equal or comparable results with a pair of dark entities, but with one unifying mechanism.

The discussion of this work is not meant to weaken the main thesis, but to place it into a broader context. GSC shows that a unifying dynamical field can carry the role that standard cosmology divides between two separate dark components. This in itself changes the way current cosmological data may be read.

At the same time, this work shows that numerical competition with ΛCDM is already serious enough that GSC should not be considered a marginal hypothesis. On the contrary, the results indicate that a model based on a dynamical field may be a fully-fledged cosmological candidate.

9 Conclusion

We have presented a summary publication construction of the GSC model as a dynamical-field cosmology that extends the standard ΛCDM into the early Universe and replaces dark matter and dark energy with a unified field. Numerical runs, including decisive MegaALL tests, show that GSC is comparable to ΛCDM in the hardest combined tests, while at the same time providing a deeper physical explanation without the need for separate dark phenomena.

This is precisely where the main strength of the model lies. ΛCDM keeps pace with GSC only because it uses dark matter and dark energy. GSC establishes itself without them. This makes it not only an alternative, but also a strong candidate for a more unified cosmological framework.

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